LAB 3

Working with Git History, Stashing, and Reverting Commits

DevOps – Lab Solution Document

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Objective

This document presents the completed work for Lab 3. The lab covers advanced Git operations that manage commit history, recover earlier versions, and safely undo mistakes both locally and remotely: handling local and remote commit conflicts, resolving merge conflicts manually, managing ignored files with `.gitignore`, stashing temporary changes, resetting, amending and reverting commits, using force push with caution, running Gitea through Docker Compose in GitHub Codespaces, and publishing a site with GitHub Pages. Each section below gives a short explanation of the task followed by the captioned screenshot evidence in the order of the lab instructions.

Task 1: Handling Local and Remote Commit Conflicts (Pull vs Pull --rebase)

Explanation. This task shows what happens when both the local repository and the GitHub remote receive new commits. A line was first added to README.md directly on GitHub, and a different line was then committed locally. The local push was rejected because the remote had work that was not present locally. The divergence was resolved first with `git pull --no-rebase` (merge strategy) and then with `git pull --rebase` (linear history).

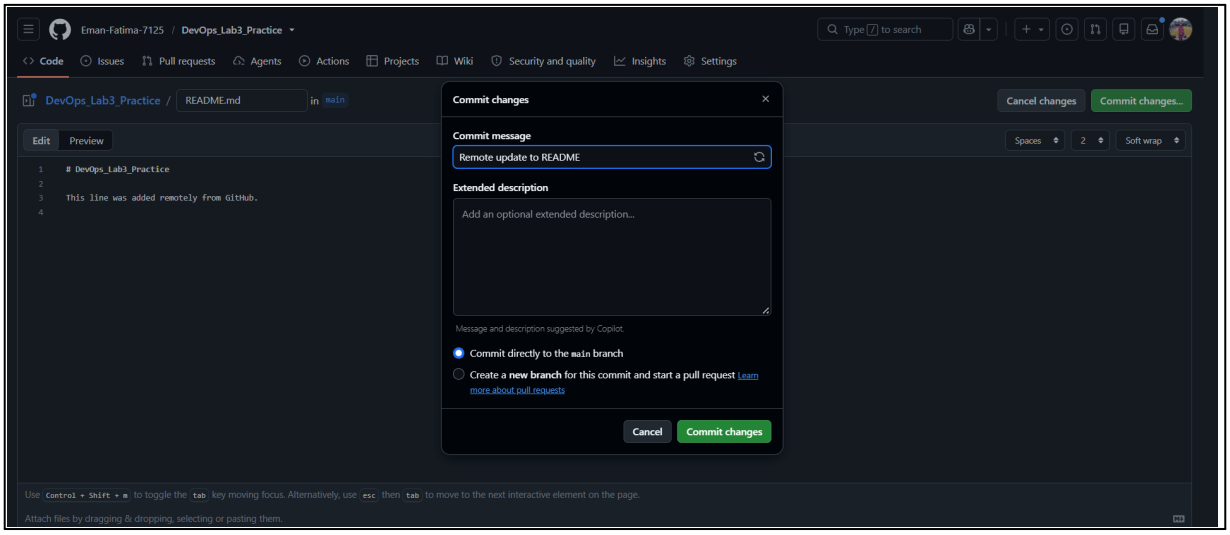


Figure 1: *README.md edited directly on GitHub and committed with the message "Remote update to README".*

```
eman-fatima-7125@ubuntu:~/DevOps_Lab3_Practice$ cat README.md
<<<<<<< HEAD
This line was added locally
=======
# DevOps_Lab3_Practice
This line was added remotely from GitHub.
>>>>>>> d1fb19303bd7b96ef85d1d7f030ab0a2111e6e4f
eman-fatima-7125@ubuntu:~/DevOps_Lab3_Practice$ eman-fatima-7125@ubuntu:~/DevOps_Lab3_Practieman-fatima-7125@ubuntu:~
/DevOps_Lab3_Practice$ cat > README.md <<'EOF'
> # DevOps_Lab3_Practice
>
> This line was added locally
> This line was added remotely from GitHub.
> EOF
eman-fatima-7125@ubuntu:~/DevOps_Lab3_Practice$ cat README.md
# DevOps_Lab3_Practice

This line was added locally
This line was added remotely from GitHub.
eman-fatima-7125@ubuntu:~/DevOps_Lab3_Practice$ git add README.md
eman-fatima-7125@ubuntu:~/DevOps_Lab3_Practice$ git commit -m "Merge local and remote histories"
[main 7587a4c] Merge local and remote histories
eman-fatima-7125@ubuntu:~/DevOps_Lab3_Practice$ git push -u origin main
Enumerating objects: 21, done.
Counting objects: 100% (21/21), done.
Delta compression using up to 2 threads
Compressing objects: 100% (11/11), done.
```

Figure 2: *Local README.md changes staged and committed, then merged with the remote content before pushing.*

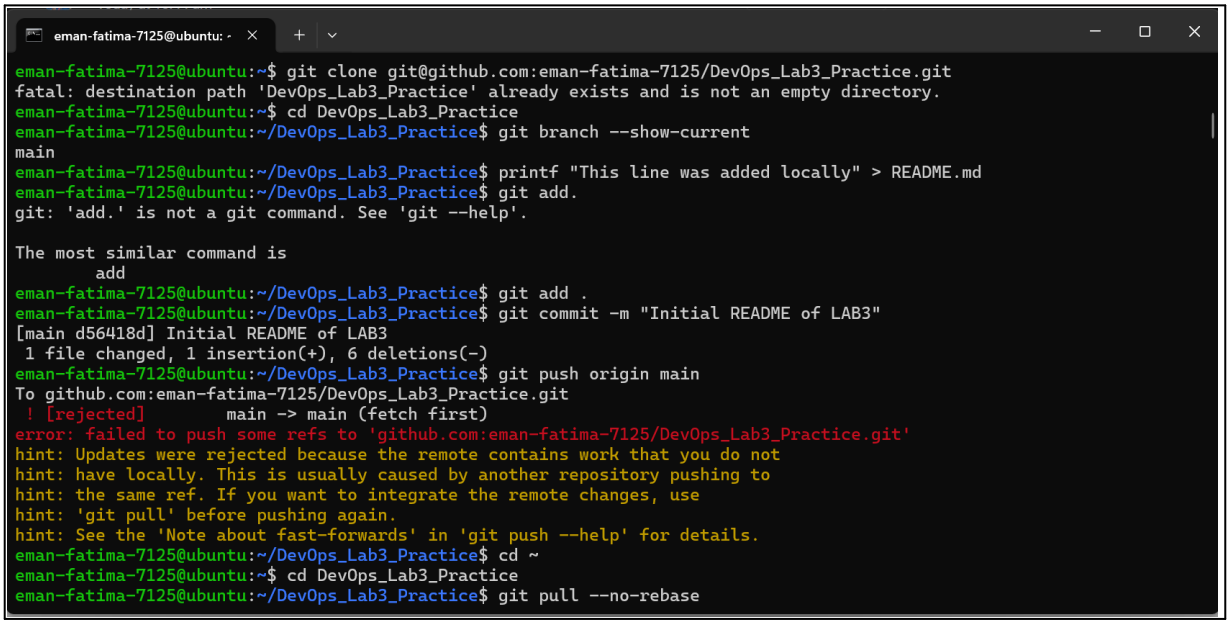


Figure 3: *Terminal output showing the local commit and the rejected* `git push origin main` *(fetch first) because the remote contained commits not present locally.*

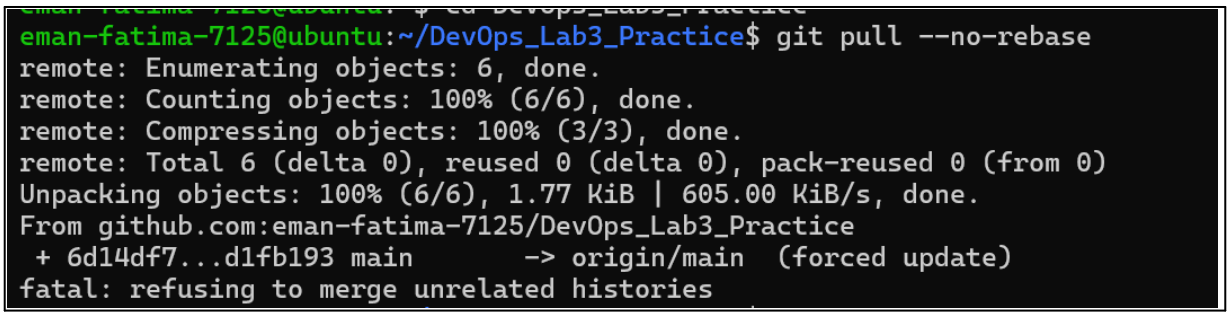


Figure 4: *Running* `git pull --no-rebase` *to integrate the remote commits; the output shows how Git responded when merging the two histories.*

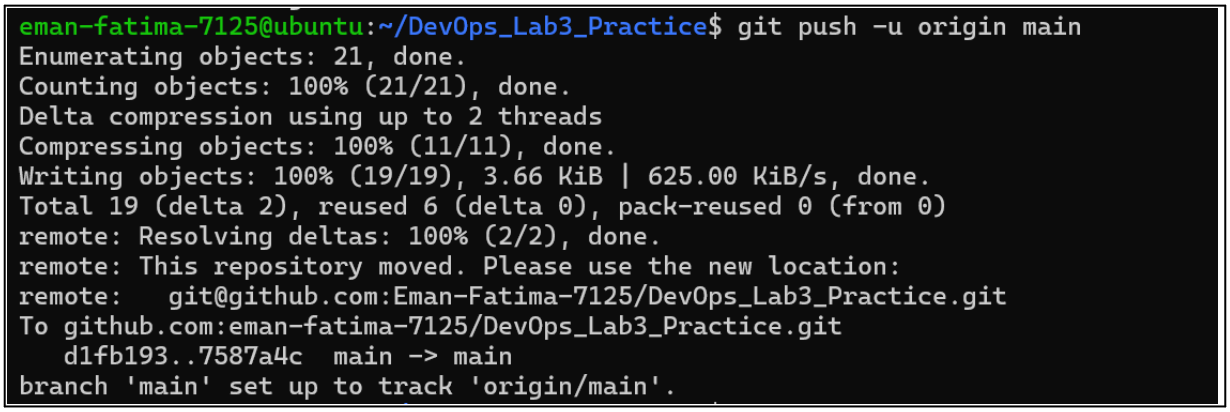


Figure 5: *Successful* `git push -u origin main` *after the merge, updating the remote branch and setting up upstream tracking.*

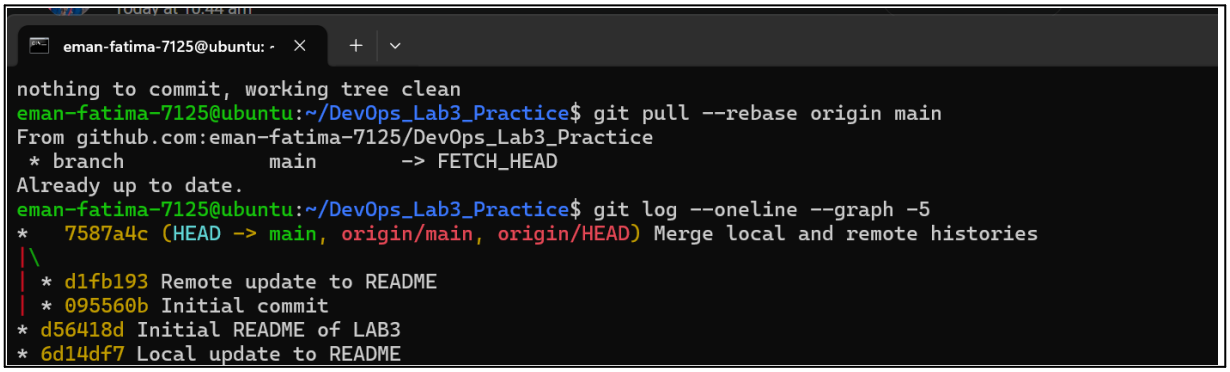


Figure 6: `git pull --rebase origin main` *followed by* `git log --oneline --graph`*, displaying the commit history of the branch.*

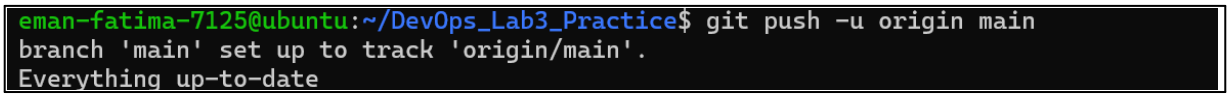


Figure 7: *Final push to* `origin/main` *after the rebase, confirming the branch is up to date.*

Task 2: Creating and Resolving Merge Conflicts Manually

Explanation. A merge conflict occurs when the same line of a file is changed differently in two places. In this task a line of README.md was changed on GitHub and the same line was changed locally. The push was rejected, `git pull --rebase` was used to bring in the remote change, the conflicting file was edited by hand to keep the intended content, and the rebase was continued before pushing the result.

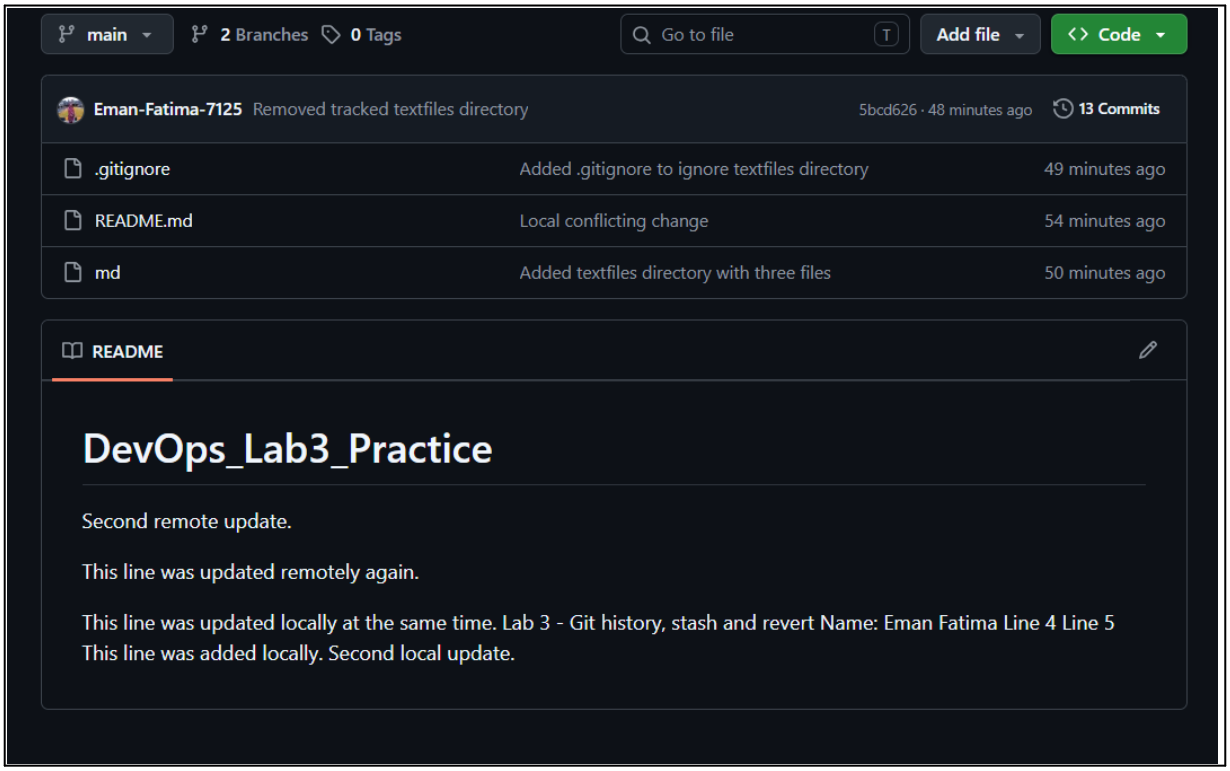


Figure 8: *Repository on GitHub after the conflicting remote change to README.md was committed.*

```
eman-fatima-7125@ubuntu:~/DevOps_Lab3_Practice$ sed -i 's/^This line was added remotely from Github\./This line was u
pdated locally at the same time./' README.md
eman-fatima-7125@ubuntu:~/DevOps_Lab3_Practice$ git diff
eman-fatima-7125@ubuntu:~/DevOps_Lab3_Practice$ git diff diff --git a/README.md b/README.md
fatal: ambiguous argument 'diff': unknown revision or path not in the working tree.
Use '--' to separate paths from revisions, like this:
'git <command> [<revision>...] -- [<file>...]'
eman-fatima-7125@ubuntu:~/DevOps_Lab3_Practice$ git diff
eman-fatima-7125@ubuntu:~/DevOps_Lab3_Practice$ git add README.md
eman-fatima-7125@ubuntu:~/DevOps_Lab3_Practice$ git commit -m "Local conflicting change"
On branch main
Your branch is up to date with 'origin/main'.

nothing to commit, working tree clean
eman-fatima-7125@ubuntu:~/DevOps_Lab3_Practice$ git push -u origin main
```

Figure 9: *Local edit of the same README.md line using* `sed`*, followed by staging, committing and an attempted push.*

```
eman-fatima-7125@ubuntu:~/DevOps_Lab3_Practice$ git add README.md
eman-fatima-7125@ubuntu:~/DevOps_Lab3_Practice$ git commit -m "Local conflicting change"
On branch main
Your branch is up to date with 'origin/main'.

nothing to commit, working tree clean
```

Figure 10: *Staging README.md and committing with the message "Local conflicting change".*

```
eman-fatima-7125@ubuntu:~/DevOps_Lab3_Practice$ git push -u origin main
branch 'main' set up to track 'origin/main'.
Everything up-to-date
```

Figure 11: *Push of the local branch to* `origin main` *using* `git push -u origin main`.

```
eman-fatima-7125@ubuntu:~/DevOps_Lab3_Practice$ git pull --rebase origin main
From github.com:eman-fatima-7125/DevOps_Lab3_Practice
 * branch            main       -> FETCH_HEAD
Already up to date.
```

Figure 12: `git pull --rebase origin main` *run to fetch the remote changes and replay the local commits on top of them.*

```
eman-fatima-7125@ubuntu:~/DevOps_Lab3_Practice$ cat -n README.md
     1  # DevOps_Lab3_Practice
     2
     3  Second remote update.
     4
     5  This line was updated remotely again.
     6
     7  This line was updated locally at the same time.
     8  Lab 3 - Git history, stash and revert
     9  Name: Eman Fatima
    10  Line 4
    11  Line 5
    12  This line was added locally.
    13  Second local update.
```

Figure 13: *README.md displayed with* `cat -n` *after manual editing, with no conflict markers remaining.*

```
eman-fatima-7125@ubuntu:~/DevOps_Lab3_Practice$ git add README.md
eman-fatima-7125@ubuntu:~/DevOps_Lab3_Practice$ git rebase --continue
fatal: no rebase in progress
```

Figure 14: *Marking the file as resolved with* `git add README.md` *and running* `git rebase --continue`.

```
eman-fatima-7125@ubuntu:~/DevOps_Lab3_Practice$ git push -u origin main
branch 'main' set up to track 'origin/main'.
Everything up-to-date
```

Figure 15: *Final* `git push -u origin main` *after the conflict was resolved; the branch is up to date.*

Task 3: Managing Ignored Files with .gitignore and Removing Tracked Files

Explanation. A `.gitignore` file prevents new files from being tracked, but it does not affect files that Git already tracks. In this task a `textfiles` directory with three files was committed and pushed, a `.gitignore` rule was added, and the directory was then removed from the index with `git rm -r --cached` so that it was deleted from the remote while remaining on the local disk.

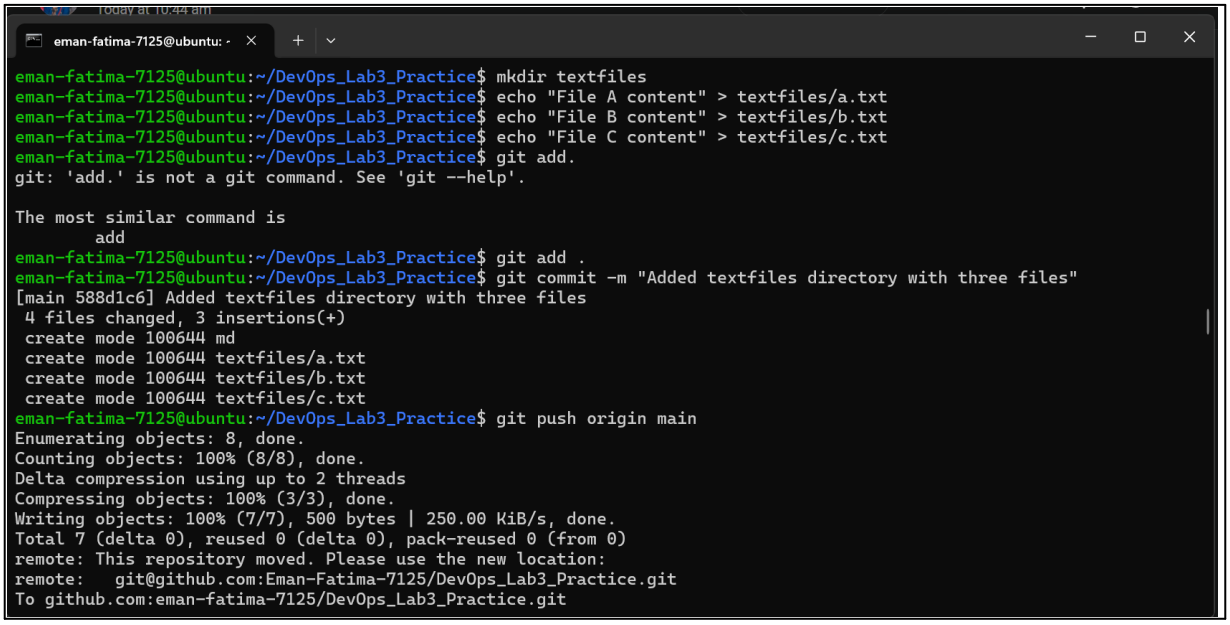


Figure 16: *Creating the* `textfiles` *directory with three files, committing them and pushing to GitHub.*

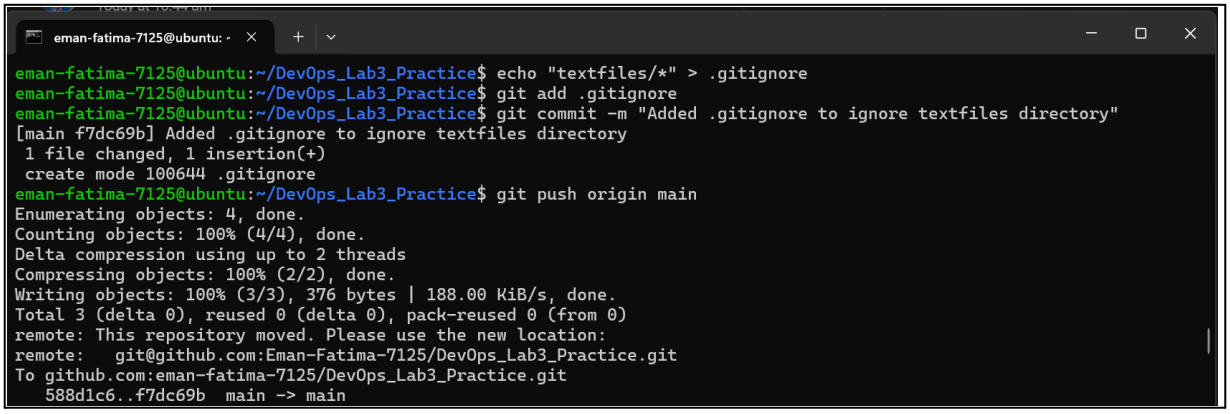


Figure 17: *Creating* `.gitignore` *with the rule* `textfiles/*`*, committing it and pushing it to the remote.*

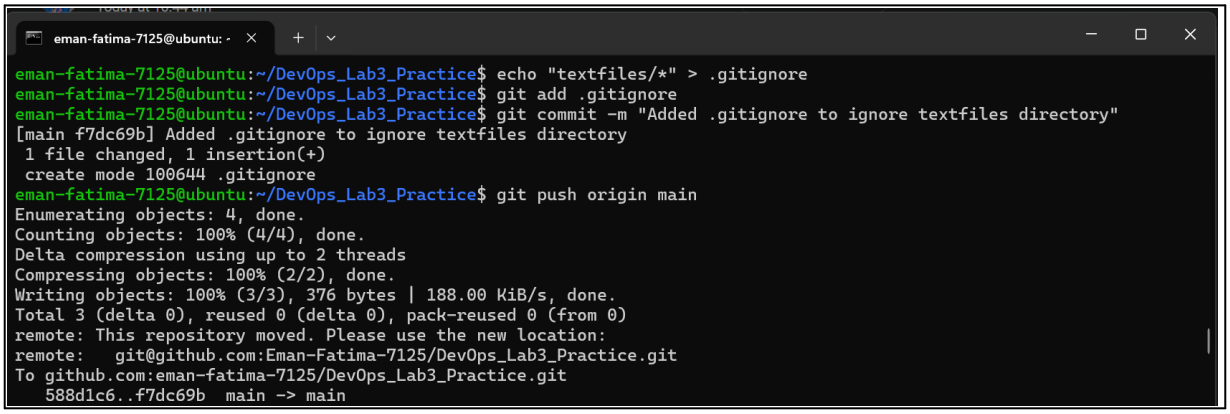


Figure 18: *Terminal showing the* `.gitignore` *commit and push, after which the previously tracked files remain in the remote repository.*

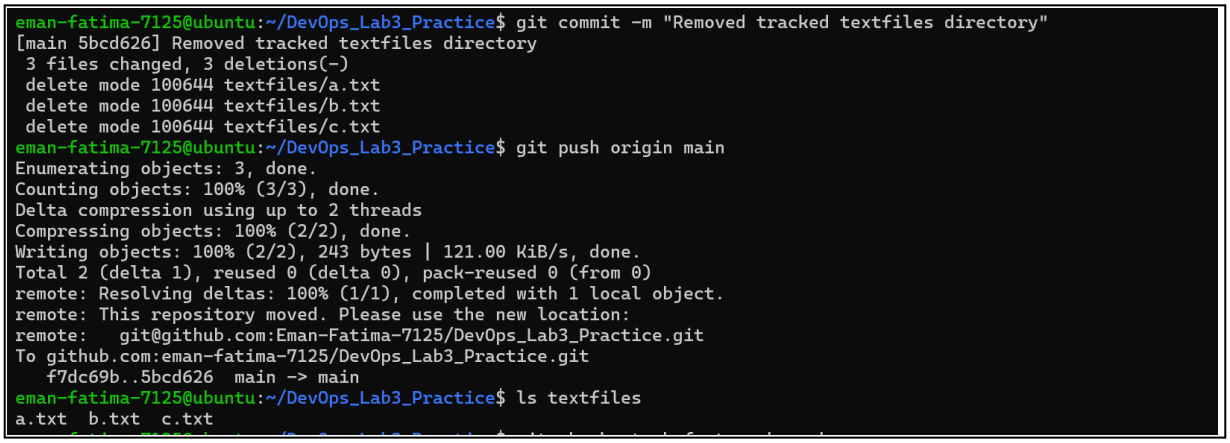


Figure 19: *Committing the removal of the tracked files and pushing;* `ls textfiles` *confirms a.txt, b.txt and c.txt still exist locally.*

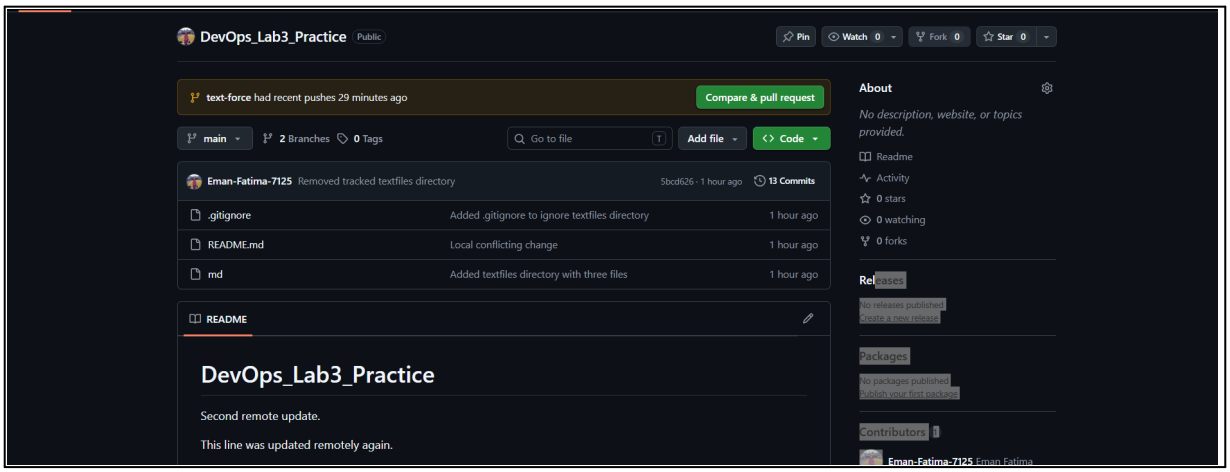


Figure 20: *GitHub repository page after the push, showing the repository contents with the tracked* `textfiles` *files removed from the remote.*

Task 4: Create Temporary Changes and Use git stash

Explanation. `git stash` saves uncommitted work so that branches can be switched safely. A feature branch was created and committed to, further uncommitted edits were made, and an attempt to switch to `main` was blocked by Git. The changes were stashed, the branch switch succeeded, and the stashed work was restored with `git stash pop`.

```
eman-fatima-7125@ubuntu:~/DevOps_Lab3_Practice$ git checkout -b feature-branch
Switched to a new branch 'feature-branch'
eman-fatima-7125@ubuntu:~/DevOps_Lab3_Practice$ echo "Feature branch line 1" >> README.md
eman-fatima-7125@ubuntu:~/DevOps_Lab3_Practice$ echo "Feature branch line 2" >> README.md
eman-fatima-7125@ubuntu:~/DevOps_Lab3_Practice$ git add README.md
eman-fatima-7125@ubuntu:~/DevOps_Lab3_Practice$ git commit -m "Changes the README.md file"
[feature-branch 930e0e9] Changes the README.md file
 1 file changed, 2 insertions(+)
eman-fatima-7125@ubuntu:~/DevOps_Lab3_Practice$ git log --oneline -3
930e0e9 (HEAD -> feature-branch) Changes the README.md file
5bcd626 (origin/main, origin/HEAD, main) Removed tracked textfiles directory
f7dc69b Added .gitignore to ignore textfiles directory
eman-fatima-7125@ubuntu:~/DevOps_Lab3_Practice$ echo "uncommitted change line 1" >> README.md
eman-fatima-7125@ubuntu:~/DevOps_Lab3_Practice$ echo "Uncommitted change line 2" >> README.md
eman-fatima-7125@ubuntu:~/DevOps_Lab3_Practice$ git checkout main
error: Your local changes to the following files would be overwritten by checkout:
        README.md
Please commit your changes or stash them before you switch branches.
Aborting
```

Figure 21: *Creating* `feature-branch`*, committing README.md changes, and adding uncommitted edits, after which checkout to main is refused.*

```
eman-fatima-7125@ubuntu:~/DevOps_Lab3_Practice$ echo "uncommitted change line 1" >> README.md
eman-fatima-7125@ubuntu:~/DevOps_Lab3_Practice$ echo "Uncommitted change line 2" >> README.md
eman-fatima-7125@ubuntu:~/DevOps_Lab3_Practice$ git checkout main
error: Your local changes to the following files would be overwritten by checkout:
        README.md
Please commit your changes or stash them before you switch branches.
Aborting
```

Figure 22: *Error message when running* `git checkout main` *with uncommitted changes that would be overwritten.*

```
eman-fatima-7125@ubuntu:~/DevOps_Lab3_Practice$ git stash
Saved working directory and index state WIP on feature-branch: 930e0e9 Changes the README.md file
```

Figure 23: *Running* `git stash` *to save the working directory and index state.*

```
eman-fatima-7125@ubuntu:~/DevOps_Lab3_Practice$ git checkout feature-branch
Switched to branch 'feature-branch'
```

Figure 24: *Switching branches again after stashing the changes.*

```
eman-fatima-7125@ubuntu:~/DevOps_Lab3_Practice$ git checkout feature -branch
fatal: 'feature' is not a commit and a branch 'ranch' cannot be created from it
eman-fatima-7125@ubuntu:~/DevOps_Lab3_Practice$ git checkout feature-branch
Switched to branch 'feature-branch'
```

Figure 25: *Returning to* `feature-branch` *after the stash.*

```
eman-fatima-7125@ubuntu:~/DevOps_Lab3_Practice$ git status
On branch feature-branch
nothing to commit, working tree clean
```

Figure 26: `git status` *reporting "nothing to commit, working tree clean" on* `feature-branch`*.*

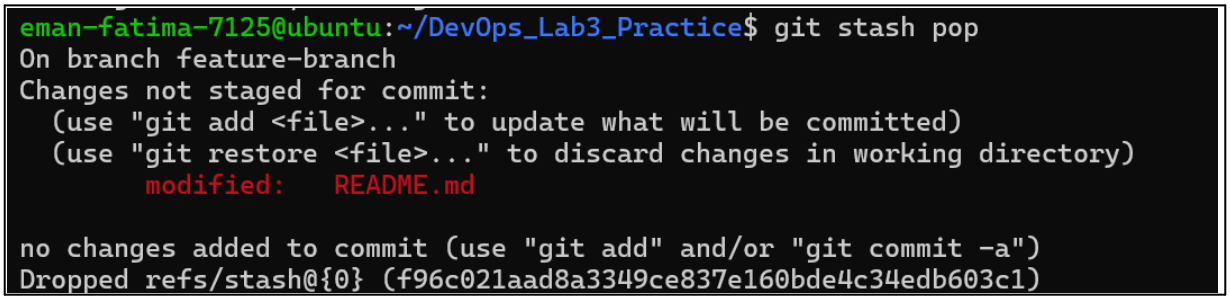


Figure 27: `git stash pop` *restoring the stashed modifications to README.md and dropping the stash entry.*

Task 5: Checkout a Specific Commit Using git log

Explanation. Any earlier version of a project can be inspected by checking out its commit hash. This places the repository in a detached HEAD state in which new commits belong to no branch. The commit history was listed with `git log --oneline`, an earlier commit was checked out, and the repository was returned to the main branch.

```
eman-fatima-7125@ubuntu:~/DevOps_Lab3_Practice$ git log --oneline
930e0e9 (HEAD -> feature-branch) Changes the README.md file
5bcd626 (origin/main, origin/HEAD, main) Removed tracked textfiles directory
f7dc69b Added .gitignore to ignore textfiles directory
588d1c6 Added textfiles directory with three files
323f8fe Local conflicting change
7587a4c Merge local and remote histories
d56418d Initial README of LAB3
d1fb193 Remote update to README
095560b Initial commit
6d14df7 Local update to README
e20d361 Setup README for Lab 3
0dd5566 Remote update to README
79ef3e0 Update README.md
9068f66 Initial commit
```

Figure 28: *Commit history displayed with* `git log --oneline`*, listing the hashes available for checkout.*

```
eman-fatima-7125@ubuntu:~/DevOps_Lab3_Practice$ git checkout 181168f
error: pathspec '181168f' did not match any file(s) known to git
eman-fatima-7125@ubuntu:~/DevOps_Lab3_Practice$ git checkout main
error: Your local changes to the following files would be overwritten by checkout:
        README.md
Please commit your changes or stash them before you switch branches.
Aborting
```

Figure 29: *Attempt to check out a specific commit hash and the resulting Git output.*

```
eman-fatima-7125@ubuntu:~/DevOps_Lab3_Practice$ git checkout main
error: Your local changes to the following files would be overwritten by checkout:
        README.md
Please commit your changes or stash them before you switch branches.
Aborting
```

Figure 30: *Attempting to return to* `main` *with* `git checkout main`*.*

Task 6: Resetting Commits (Soft vs Hard Reset)

Explanation. A soft reset moves the branch pointer back but keeps the changes staged, while a hard reset also discards the changes from the working directory. Two test commits were created, the history and file content were verified, a soft reset was performed (changes retained), the work was committed again, and finally a hard reset removed the latest commit and its changes.

```
eman-fatima-7125@ubuntu:~/DevOps_Lab3_Practice$ echo "Test line 1" >>README.md
eman-fatima-7125@ubuntu:~/DevOps_Lab3_Practice$ git add .
eman-fatima-7125@ubuntu:~/DevOps_Lab3_Practice$ git commit -m "Added test line"
[feature-branch 109fc74] Added test line
 1 file changed, 3 insertions(+)
```

Figure 31: *First test commit, "Added test line", created on* `feature-branch`.

```
eman-fatima-7125@ubuntu:~/DevOps_Lab3_Practice$ echo "Test line 2" >> README.md
eman-fatima-7125@ubuntu:~/DevOps_Lab3_Practice$ git add .
eman-fatima-7125@ubuntu:~/DevOps_Lab3_Practice$ git commit -m "Second test commit"
[feature-branch 3b4d1aa] Second test commit
 1 file changed, 1 insertion(+)
```

Figure 32: *Second test commit, "Second test commit", created after adding another line.*

```
eman-fatima-7125@ubuntu:~/DevOps_Lab3_Practice$ git log --oneline -5
3b4d1aa (HEAD -> feature-branch) Second test commit
109fc74 Added test line
930e0e9 Changes the README.md file
5bcd626 (origin/main, origin/HEAD, main) Removed tracked textfiles directory
f7dc69b Added .gitignore to ignore textfiles directory
```

Figure 33: `git log --oneline -5` *showing both test commits before any reset.*

```
eman-fatima-7125@ubuntu:~/DevOps_Lab3_Practice$ cat README.md
# DevOps_Lab3_Practice

Second remote update.

This line was updated remotely again.

This line was updated locally at the same time.
Lab 3 - Git history, stash and revert
Name: Eman Fatima
Line 4
Line 5
This line was added locally.
Second local update.
Feature branch line 1
Feature branch line 2
uncommitted change line 1
Uncommitted change line 2
Test line 1
Test line 2
```

Figure 34: *README.md contents showing both "Test line 1" and "Test line 2" before the reset.*

```
eman-fatima-7125@ubuntu:~/DevOps_Lab3_Practice$ git reset --soft HEAD~1
eman-fatima-7125@ubuntu:~/DevOps_Lab3_Practice$ git status
On branch feature-branch
Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
        modified:   README.md
```

Figure 35: `git reset --soft HEAD~1` *followed by* `git status`*, showing README.md as a staged change.*

```
eman-fatima-7125@ubuntu:~/DevOps_Lab3_Practice$ git reset --soft HEAD~1
eman-fatima-7125@ubuntu:~/DevOps_Lab3_Practice$ git log --oneline -5
930e0e9 (HEAD -> feature-branch) Changes the README.md file
5bcd626 (origin/main, origin/HEAD, main) Removed tracked textfiles directory
f7dc69b Added .gitignore to ignore textfiles directory
588d1c6 Added textfiles directory with three files
323f8fe Local conflicting change
```

Figure 36: *Commit history after the soft reset, with the latest commit removed from the log.*

```
eman-fatima-7125@ubuntu:~/DevOps_Lab3_Practice$ cat README.md
# DevOps_Lab3_Practice

Second remote update.

This line was updated remotely again.

This line was updated locally at the same time.
Lab 3 - Git history, stash and revert
Name: Eman Fatima
Line 4
Line 5
This line was added locally.
Second local update.
Feature branch line 1
Feature branch line 2
uncommitted change line 1
Uncommitted change line 2
Test line 1
Test line 2
eman-fatima-7125@ubuntu:~/DevOps_Lab3_Practice$ git status
On branch feature-branch
Changes to be committed:
  (use "git restore --staged <file>..." to unstage)
        modified:   README.md
```

Figure 37: *README.md after the soft reset: both added lines are still present and the file is staged for commit.*

```
eman-fatima-7125@ubuntu:~/DevOps_Lab3_Practice$ git commit -m "Second Test commit"
[feature-branch 89c1cc0] Second Test commit
 1 file changed, 4 insertions(+)
```

Figure 38: *Re-committing the staged changes with the message "Second Test commit".*

```
eman-fatima-7125@ubuntu:~/DevOps_Lab3_Practice$ git reset --hard HEAD~1
HEAD is now at 930e0e9 Changes the README.md file
```

Figure 39: `git reset --hard HEAD~1` *moving HEAD back to the earlier commit.*

```
eman-fatima-7125@ubuntu:~/DevOps_Lab3_Practice$ git log --oneline -5
930e0e9 (HEAD -> feature-branch) Changes the README.md file
5bcd626 (origin/main, origin/HEAD, main) Removed tracked textfiles directory
f7dc69b Added .gitignore to ignore textfiles directory
588d1c6 Added textfiles directory with three files
323f8fe Local conflicting change
```

Figure 40: *Commit history after the hard reset, showing HEAD at the earlier commit.*

```
eman-fatima-7125@ubuntu:~/DevOps_Lab3_Practice$ cat README.md
# DevOps_Lab3_Practice

Second remote update.

This line was updated remotely again.

This line was updated locally at the same time.
Lab 3 - Git history, stash and revert
Name: Eman Fatima
Line 4
Line 5
This line was added locally.
Second local update.
Feature branch line 1
Feature branch line 2
```

Figure 41: *README.md after the hard reset: the lines added by the discarded commit are no longer present.*

Task 7: Amending the Last Commit

Explanation. `git commit --amend` rewrites the most recent commit so that a forgotten change or a poor message can be corrected without creating an additional commit. A small change was committed, and the history was then inspected to show the commit that is amended.

```
eman-fatima-7125@ubuntu:~/DevOps_Lab3_Practice$ echo "log fix" >> README.md
eman-fatima-7125@ubuntu:~/DevOps_Lab3_Practice$ git add .
eman-fatima-7125@ubuntu:~/DevOps_Lab3_Practice$ git commit -m "Fix log message"
[feature-branch bfa2e55] Fix log message
 1 file changed, 1 insertion(+)
eman-fatima-7125@ubuntu:~/DevOps_Lab3_Practice$ git log --oneline -2
bfa2e55 (HEAD -> feature-branch) Fix log message
930e0e9 Changes the README.md file
```

Figure 42: *A change appended to README.md, staged and committed with the message "Fix log message".*

```
eman-fatima-7125@ubuntu:~/DevOps_Lab3_Practice$ git log --oneline -3
bfa2e55 (HEAD -> feature-branch) Fix log message
930e0e9 Changes the README.md file
5bcd626 (origin/main, origin/HEAD, main) Removed tracked textfiles directory
```

Figure 43: `git log --oneline -3` *showing the latest commit at the top of* `feature-branch`*.*

Task 8: Reverting a Commit (Safe Undo on Remote Branch)

Explanation. `git revert` undoes the effect of an earlier commit by adding a new commit, so shared history is not rewritten. A temporary file was committed, the revert was attempted using the commit hash from the log, and the result was pushed to the remote.

```
eman-fatima-7125@ubuntu:~/DevOps_Lab3_Practice$ git commit -m "Added temporary text file"
[feature-branch 534fb48] Added temporary text file
 1 file changed, 1 insertion(+)
 create mode 100644 temp.txt
eman-fatima-7125@ubuntu:~/DevOps_Lab3_Practice$ git push origin main
Everything up-to-date
eman-fatima-7125@ubuntu:~/DevOps_Lab3_Practice$ git log --oneline -3
534fb48 (HEAD -> feature-branch) Added temporary text file
bfa2e55 Fix log message
930e0e9 Changes the README.md file
```

Figure 44: *Committing* `temp.txt` *with the message "Added temporary text file", pushing, and viewing the log.*

```
eman-fatima-7125@ubuntu:~/DevOps_Lab3_Practice$ git log --oneline -3
534fb48 (HEAD -> feature-branch) Added temporary text file
bfa2e55 Fix log message
930e0e9 Changes the README.md file
eman-fatima-7125@ubuntu:~/DevOps_Lab3_Practice$ git revert --no-edir d1efc00
fatal: bad revision 'd1efc00'
eman-fatima-7125@ubuntu:~/DevOps_Lab3_Practice$ git log --oneline -4
534fb48 (HEAD -> feature-branch) Added temporary text file
bfa2e55 Fix log message
930e0e9 Changes the README.md file
5bcd626 (origin/main, origin/HEAD, main) Removed tracked textfiles directory
```

Figure 45: *Viewing the history and running* `git revert` *on the selected commit.*

```
eman-fatima-7125@ubuntu:~/DevOps_Lab3_Practice$ git push origin main
Everything up-to-date
```

Figure 46: *Pushing to* `origin main` *after the revert step.*

Task 9: Force Push (With Caution)

Explanation. Force pushing overwrites remote history and must only be used on a personal branch. A new branch was created and pushed, the last commit was removed locally with a hard reset, a normal push was rejected because the local branch was behind its remote counterpart, and `--force` was then used to overwrite the remote branch.

```
eman-fatima-7125@ubuntu:~/DevOps_Lab3_Practice$ git checkout -b text-force
Switched to a new branch 'text-force'
```

Figure 47: *Creating the new branch* `text-force` *with* `git checkout -b`.

```
eman-fatima-7125@ubuntu:~/DevOps_Lab3_Practice$ echo "force test line" >> force.txt
eman-fatima-7125@ubuntu:~/DevOps_Lab3_Practice$ git add .
eman-fatima-7125@ubuntu:~/DevOps_Lab3_Practice$ git commit -m "Force test commit"
[text-force e57a2c3] Force test commit
 1 file changed, 1 insertion(+)
 create mode 100644 force.txt
eman-fatima-7125@ubuntu:~/DevOps_Lab3_Practice$ git log --oneline -3
e57a2c3 (HEAD -> text-force) Force test commit
4656ee5 (feature-branch) Revert "Added temporary text file"
534fb48 Added temporary text file
```

Figure 48: *Creating* `force.txt` *and committing it with the message "Force test commit".*

```
eman-fatima-7125@ubuntu:~/DevOps_Lab3_Practice$ git push origin text-force
Enumerating objects: 15, done.
Counting objects: 100% (15/15), done.
Delta compression using up to 2 threads
Compressing objects: 100% (11/11), done.
Writing objects: 100% (13/13), 1.25 KiB | 212.00 KiB/s, done.
Total 13 (delta 5), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (5/5), completed with 1 local object.
remote: This repository moved. Please use the new location:
remote:   git@github.com:Eman-Fatima-7125/DevOps_Lab3_Practice.git
remote:
remote: Create a pull request for 'text-force' on GitHub by visiting:
remote:      https://github.com/Eman-Fatima-7125/DevOps_Lab3_Practice/pull/new/text-force
remote:
To github.com:eman-fatima-7125/DevOps_Lab3_Practice.git
 * [new branch]      text-force -> text-force
```

Figure 49: *Pushing the new branch to the remote repository for the first time.*

```
eman-fatima-7125@ubuntu:~/DevOps_Lab3_Practice$ git reset --hard HEAD~1
HEAD is now at 4656ee5 Revert "Added temporary text file"
eman-fatima-7125@ubuntu:~/DevOps_Lab3_Practice$ git log --oneline -3
4656ee5 (HEAD -> text-force, feature-branch) Revert "Added temporary text file"
534fb48 Added temporary text file
bfa2e55 Fix log message
eman-fatima-7125@ubuntu:~/DevOps_Lab3_Practice$ git push origin text-force
To github.com:eman-fatima-7125/DevOps_Lab3_Practice.git
 ! [rejected]        text-force -> text-force (non-fast-forward)
error: failed to push some refs to 'github.com:eman-fatima-7125/DevOps_Lab3_Practice.git'
hint: Updates were rejected because the tip of your current branch is behind
hint: its remote counterpart. If you want to integrate the remote changes,
hint: use 'git pull' before pushing again.
hint: See the 'Note about fast-forwards' in 'git push --help' for details.
```

Figure 50: `git reset --hard HEAD~1` *removing the commit locally, followed by a normal push that Git rejects as non-fast-forward.*

```
hint: See the 'Note about fast-forwards' in 'git push --help' for details.
eman-fatima-7125@ubuntu:~/DevOps_Lab3_Practice$ git push origin text-force --force
Total 0 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote: This repository moved. Please use the new location:
remote:   git@github.com:Eman-Fatima-7125/DevOps_Lab3_Practice.git
To github.com:eman-fatima-7125/DevOps_Lab3_Practice.git
```

Figure 51: *Running* `git push origin text-force --force` *to overwrite the remote branch history.*

Task 10: Running Gitea in GitHub Codespaces via Docker Compose

Explanation. In this task the Gitea repository was forked (with all branches), opened in a GitHub Codespace, switched to the `codespace` branch, and started with Docker Compose. Port 3000 was forwarded and made public so that the Gitea web installer and dashboard could be reached from the browser.

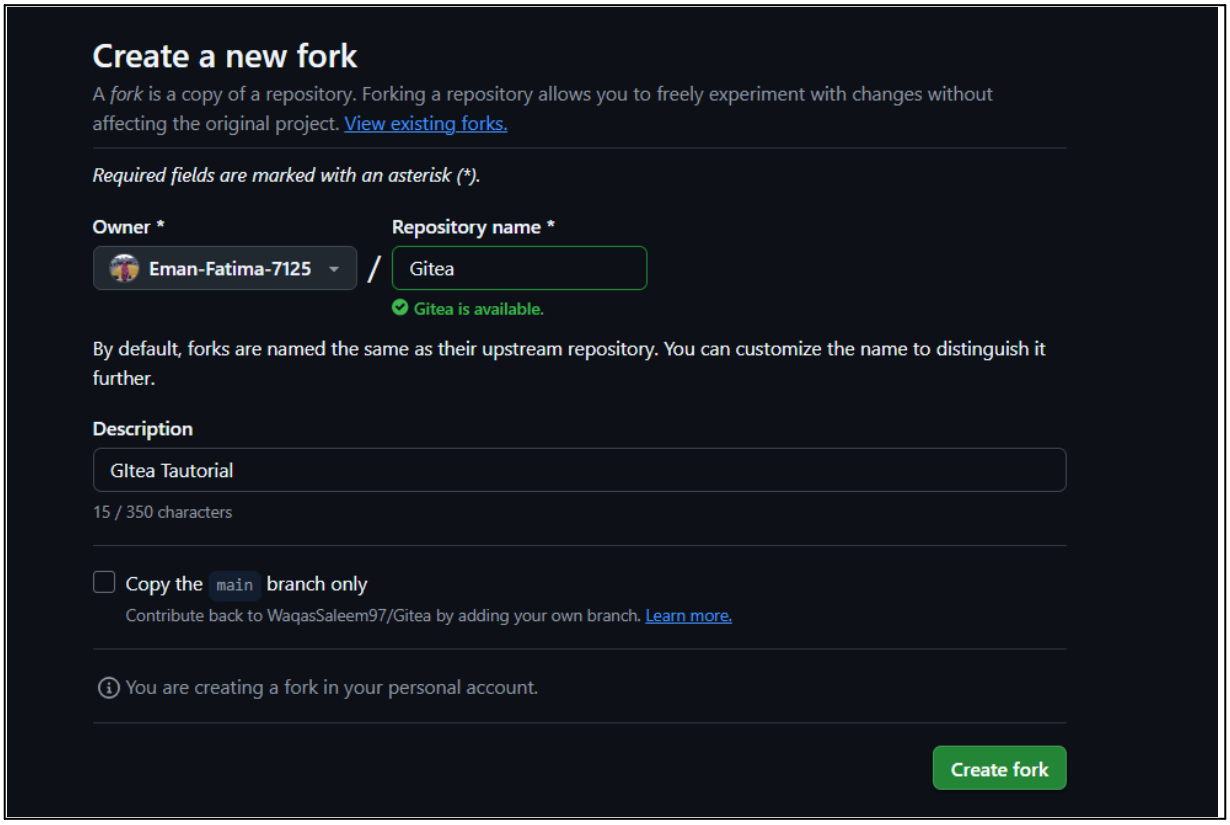


Figure 52: *Fork creation page for the Gitea repository under the account Eman-Fatima-7125, with "Copy the main branch only" unchecked.*

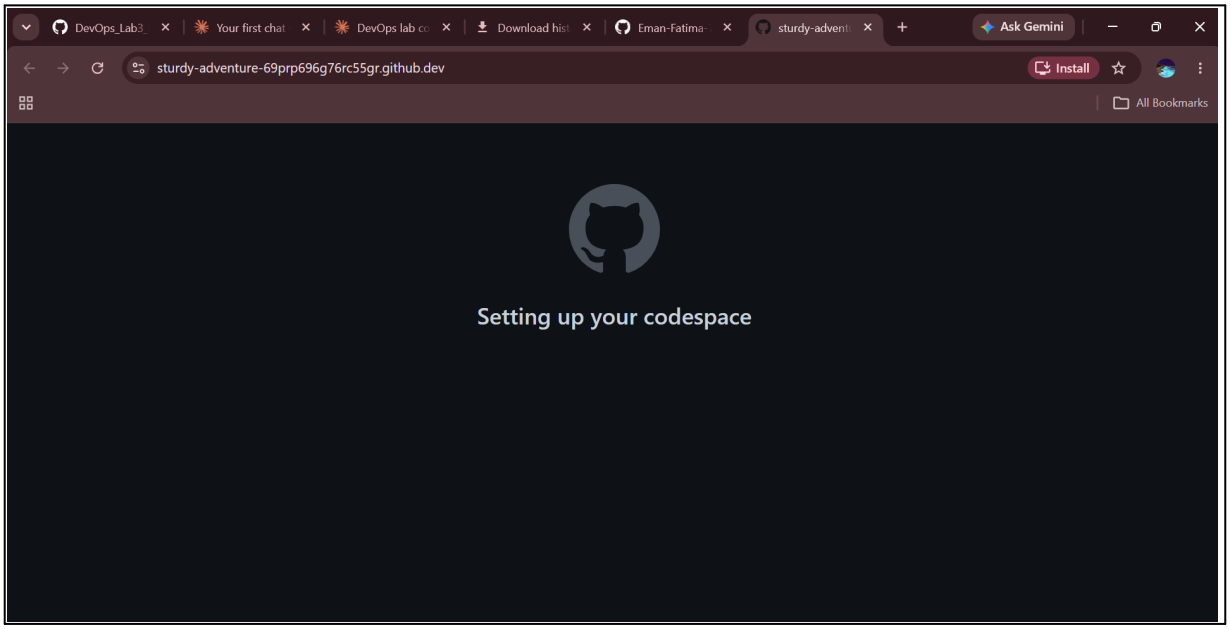


Figure 53: *The Codespace being set up for the forked repository.*

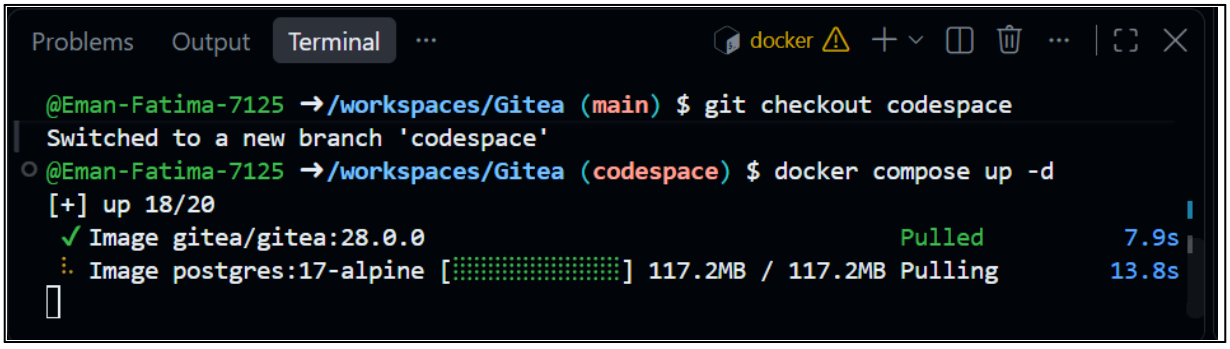


Figure 54: *Switching to the* `codespace` *branch and starting the services with* `docker compose up -d`*, with the images being pulled.*

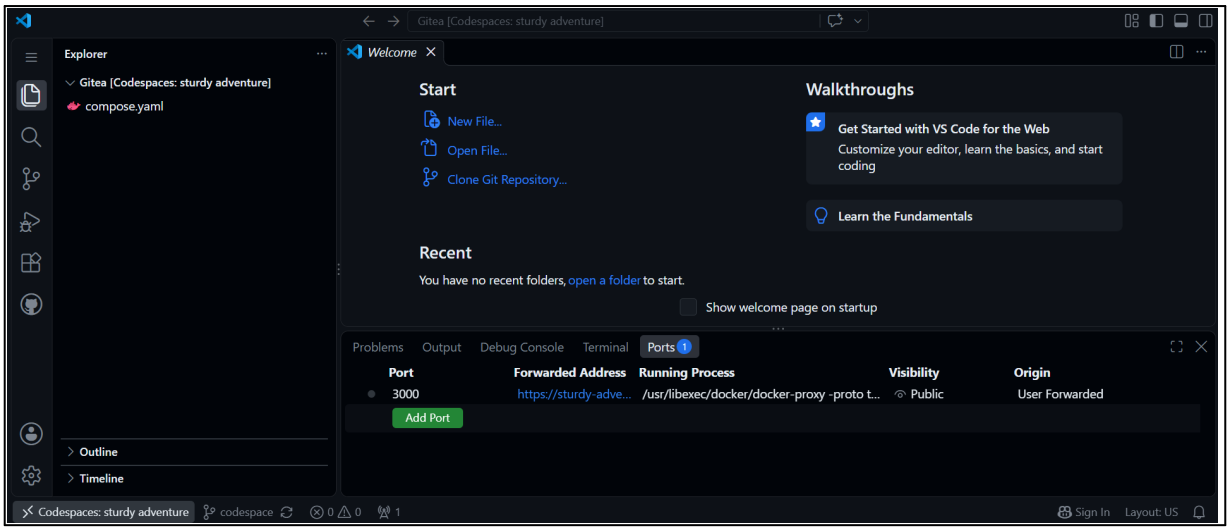


Figure 55: *Codespaces window showing the forwarded port 3000 with its visibility set to Public.*

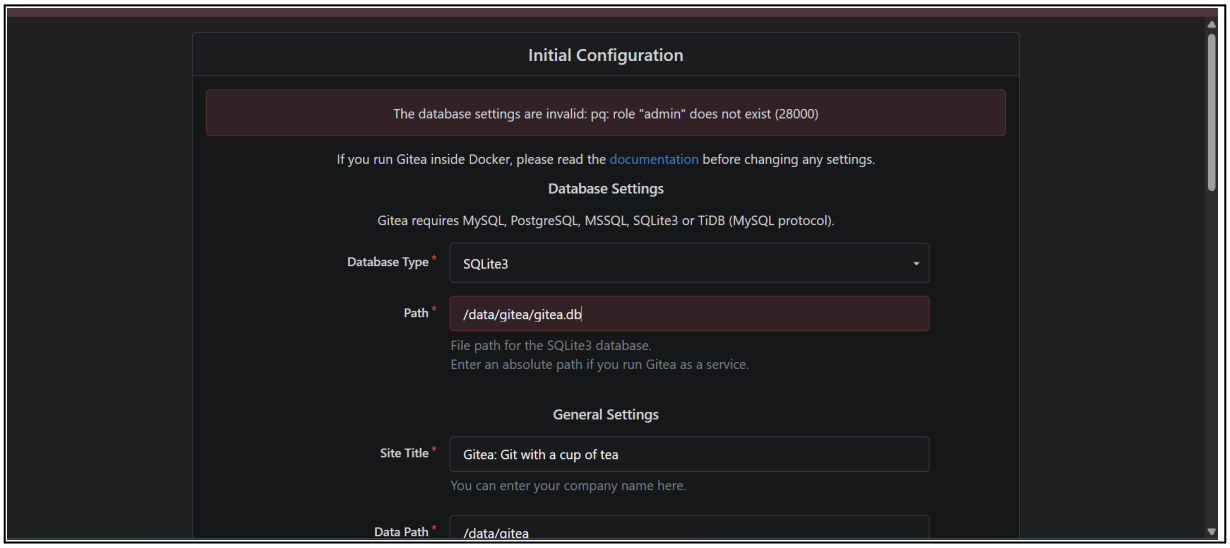


Figure 56: *Gitea Initial Configuration page with the database and general settings shown.*

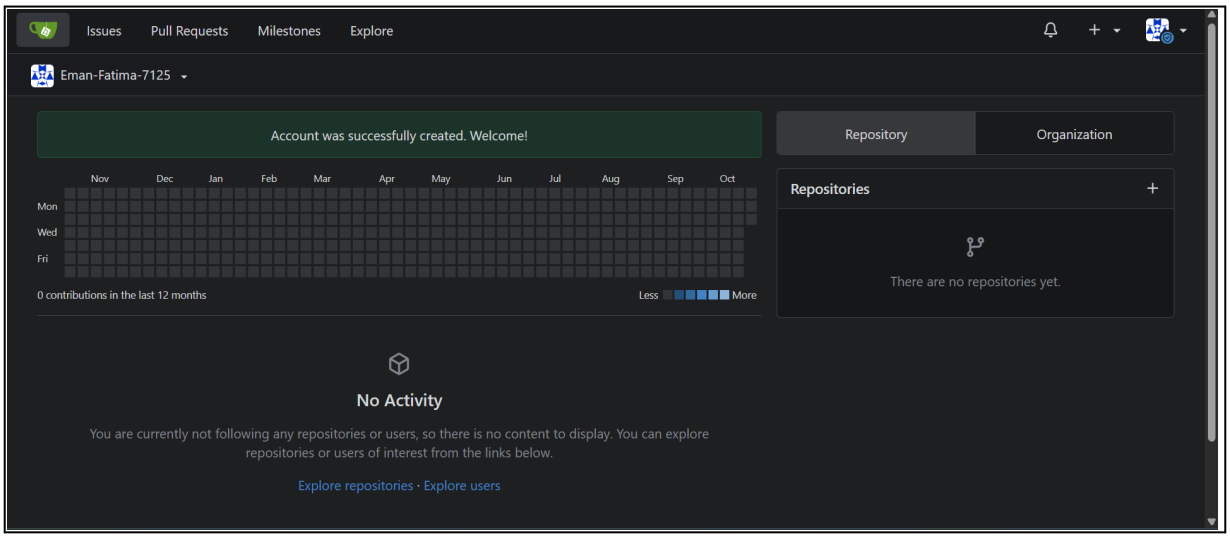


Figure 57: *Gitea dashboard after the administrator account Eman-Fatima-7125 was created and the user logged in.*

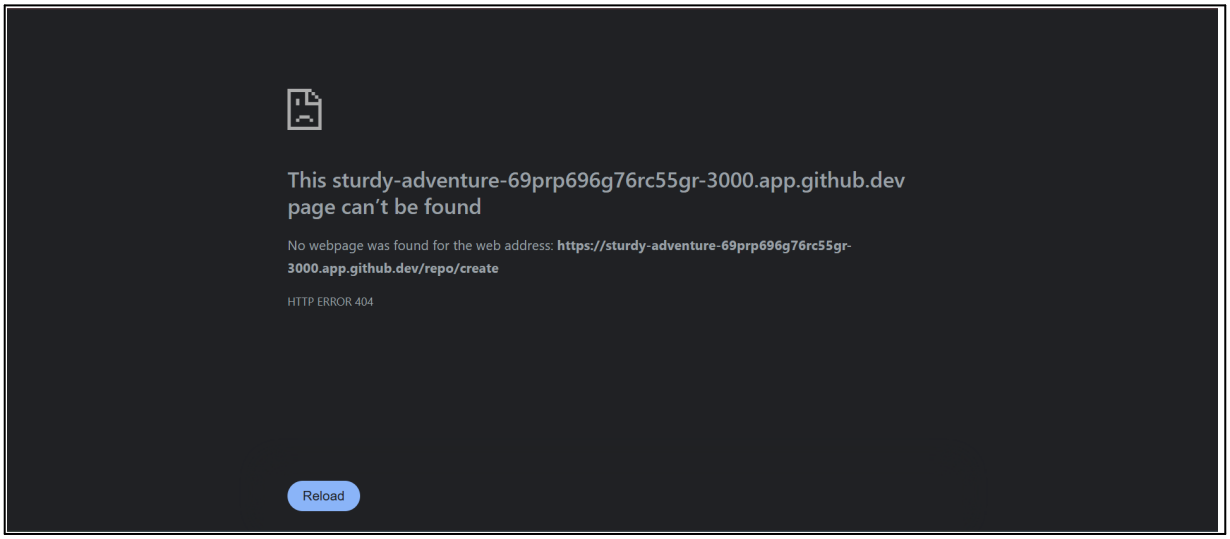


Figure 58: *Browser attempt to open the repository creation page through the forwarded Codespaces address.*

Task 11: Creating a GitHub Pages Portfolio Site

Explanation. This task publishes a static portfolio site through GitHub Pages. A public repository named `Eman-Fatima-7125.github.io` is created, the HTML/CSS files are committed and pushed to its main branch, and Pages is enabled under Settings so the site is served at `https://eman-fatima-7125.github.io`.

No screenshot evidence for this section was included in the submitted evidence set.

Exam Evaluation Questions

Explanation. The four exam questions repeat the core skills of Tasks 1 to 4: resolving local and remote divergence with merge and rebase, resolving a manual merge conflict, untracking files that are already committed, and manipulating commit history with soft and hard resets. The same command sequences demonstrated in the tasks above apply to the exam questions.

No screenshot evidence for this section was included in the submitted evidence set.